

Continuous images of Radon–Nikodým compacta need not be Radon–Nikodým

Literature answer to an open problem mentioned in arXiv:math/0610420

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Status: literature-already-answered, full negative answer. Avilés and Koszmider construct a continuous image of a zero-dimensional Radon–Nikodým compact space that is not Radon–Nikodým, explicitly resolving the closure-under-continuous-images problem mentioned by Haydon.

Source problem

On source PDF page 4, Haydon writes [1]:

“The situation is closely related to the open problem of whether every continuous image of a Radon–Nikodým compact is again a Radon–Nikodým compact.”

A compact space is Radon–Nikodým (RN) when it is fragmented by a lower semicontinuous metric; equivalently, it is homeomorphic to a weak-star compact subset of the dual of a Banach space with the Radon–Nikodým property.

Explicit later answer

The abstract of Avilés and Koszmider says that they construct such a counterexample, “solving the problem posed in the 80ties by Isaac Namioka” [2]. Their introduction restates the precise question

“Is the class of RN compact spaces closed under continuous images?”

and Theorem 1.1, on supporting PDF page 2, proves:

There exists a continuous surjection $\pi : L_0 \rightarrow L_1$ such that L_0 is a zero-dimensional RN compact space but L_1 is not RN compact.

This is a full negative answer, with the additional strength that the domain is zero-dimensional.

Provenance and scope

The supporting authors do not cite Haydon’s paper, but they explicitly state that they are solving the same named general problem and trace it to Namioka. This is therefore an explicit literature answer rather than an agent-generated counterexample.

The answer does *not* settle the separate question in the next paragraph of Haydon’s paper: whether $C(K)$ is LUR-renormable for every descriptive compact K , equivalently whether a Banach space whose dual norm is weak-star LUR must admit an equivalent LUR norm. That broader renorming question remains outside this packet.

References

- [1] R. Haydon, *Locally uniformly convex norms in Banach spaces and their duals*, arXiv:math/0610420v2 (2006), p. 4.
- [2] A. Avilés and P. Koszmider, *A continuous image of a Radon–Nikodým compact space which is not Radon–Nikodým*, arXiv:1112.4152 (2011), Theorem 1.1.